51.3: Selection for individual survival and reproductive success can explain diverse behaviors

EVOLUTION We turn now to Tinbergen's third question—how behavior enhances survival and reproduction in a population. The focus thus shifts from proximate causation—the "how" questions—to ultimate causation—the "why" questions. We'll begin by considering the activity of gathering food. Food-obtaining behavior, or **foraging**, includes not only eating but also any activities an animal uses to search for, recognize, and capture food items.

Evolution of Foraging Behavior

The fruit fly allows us to examine one way that foraging behavior might have evolved. Variation in a gene called *forager* (*for*) dictates how far *Drosophila* larvae travel when foraging. On average, larvae carrying the forR ("Rover") allele travel nearly twice as far while foraging as do larvae with the fors ("sitter") allele.

Both the forR and fors alleles are present in natural populations. What circumstances might favor one or the other allele? An answer became apparent in experiments that maintained flies at either low or high population densities for many generations. Larvae in populations kept at a low density foraged over shorter distances than those in populations kept at high density **(Figure 51.13)**. Furthermore, the fors allele increased in frequency in the low-density populations, whereas the forR allele increased in frequency in the high-density group. These changes make sense. At a low population density, short-distance foraging yields sufficient food, while long-distance foraging would result in unnecessary energy expenditure. Under crowded conditions, long-distance foraging could enable larvae to move beyond areas depleted of food. Thus, an interpretable evolutionary change in behavior occurred in the course of the experiment.

Figure 51.13

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    Evolution of foraging behavior by laboratory populations of
*Drosophila melanogaster*.
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A bar chart plots mean path length for low population density and high population density Drosophila melanogaster lineages. Mean path length is substantially higher for high population density Drosophila melanogaster lineages.

After 74 generations of living at low population density, *Drosophila* larvae (populations R1–R3) followed foraging paths significantly shorter than those of *Drosophila* larvae that had lived at high density (populations K1–K3).

INTERPRET THE DATA *What alternative hypothesis is made far less likely by having three* R *and* K *lines, rather than one of each?*

Optimal Foraging Model

To study the ultimate causation of foraging strategies, biologists sometimes apply a type of cost-benefit analysis used in economics. This idea proposes that foraging behavior is a compromise between the benefits of nutrition and the costs of obtaining food. These costs might include the energy expenditure of foraging as well as the risk of being eaten while foraging. According to this **optimal foraging model**, natural selection should favor a foraging behavior that minimizes the costs of foraging and maximizes the benefits. The **Scientific Skills Exercise** provides an example of how this model can be applied to animals in the wild.

Scientific Skills Exercise

Testing a Hypothesis with a Quantitative Model

AP Science Practices 1, 4

Do Crows Display Optimal Foraging Behavior? On islands off British Columbia, Canada, Northwestern crows (*Corvus caurinus*) search rocky tide pools for sea snails called whelks. After spotting a whelk, the crow picks it up in its beak, flies upward, and drops the whelk onto the rocks. If the drop is successful, the shell breaks and the crow can dine on the whelk's soft parts. If not, the crow flies up and drops the whelk again and again until the shell breaks. What determines how high the crow flies? If energetic considerations dominated selection for the crow's foraging behavior, the average drop height might reflect a trade-off between the cost of flying higher and the benefit of more frequent success. In this exercise you'll test how well this optimal foraging model predicts the average drop height observed in nature.

How the Experiments Were Done There were two parts to the experiment. First, the researcher measured the height of drops made by crows in the wild by referring to a marked pole erected nearby. Second, the researcher built a device that dropped a whelk onto the rocks from a fixed platform. For each whelk, he kept track of the number of drops carried out before the whelk broke open. Averaging over many trials with the device and combining the data for each platform height, he calculated a predicted total "flight" height: the height of the platform times the average number of drops required to break open the whelk.

Data from the Experiment The graph summarizes the results of the experiment, comparing the behavior of crows in the wild with the measurements using the whelk-dropping device.

A bar chart plots the average number of drops versus drop height (meters) and total flight height. Drop height preferred by crows shown as 5.23 meters.

Data are as follows:

Drop Height (meters)	Average Number of Drops approximate)	Total flight height (number of drops times drop height in meters) (approximate)
2	55	110
3	12	37
5	5	30
7	5	35
15	4	60

Data from R. Zach, Shell-dropping: Decision-making and optimal foraging in northwestern crows, *Behavior* 68:106–117 (1979).

INTERPRET THE DATA

- How does the average number of drops required to break open a whelk depend on platform height for a drop of 5 meters or less? For drops of more than 5 meters?
- Total flight height can be considered to be a measure of the total energy required to break open a whelk. Why is this value lower for a platform set at 5 meters than for one at 2 or 15 meters?
- Compare the drop height preferred by crows with the graph of total flight height for the platform drops. Are the data consistent with the hypothesis of optimal foraging? Explain.
- In testing the optimal foraging model, it was assumed that changing the height of the drop only changed the total energy required. Do you think this is a realistic limitation, or might other factors than total energy be affected by height?
- Researchers observed that the crows only gather and drop the largest whelks. What are some reasons crows might favor larger whelks?

- It turned out that the probability of a whelk breaking was the same for a whelk dropped for the first time as for an unbroken whelk dropped several times previously. If the probability of breaking instead increased, what change might you predict in the crow's behavior?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Balancing Risk and Reward

One of the most significant potential costs to a forager is risk of predation. Maximizing energy gain and minimizing energy costs are of little benefit if the behavior makes the forager a likely meal for a predator. It seems logical, therefore, that predation risk would influence foraging behavior. Such appears to be the case for the mule deer (*Odocoileus hemionus*), which lives in the mountains of western North America. Researchers found that the food available for mule deer was fairly uniform across the potential foraging areas, although somewhat lower in open, nonforested areas. In contrast, the risk of predation differed greatly; mountain lions (*Puma concolor*), the major predator, killed large numbers of mule deer at forest edges and only a small number in open areas and forest interiors.

How does mule deer foraging behavior reflect the differences in predation risk in particular areas? Mule deer feed predominantly in open areas. Thus, it appears that mule deer foraging behavior reflects the large variation in predation risk and not the smaller variation in food availability. This result underscores the point that behavior typically reflects a compromise between competing selective pressures.

Mating Behavior and Mate Choice

Just as foraging is crucial for individual survival, mating behavior and mate choice play a major role in determining reproductive success. These behaviors include seeking or attracting mates, choosing among potential mates, competing for mates, and caring for offspring.

Mating Systems and Sexual Dimorphism

There is considerable variation in patterns of sexual contact. In the context of reproduction, biologists describe differences in *mating systems*, the length and number of relationships between males and females. In some animal species, mating is *promiscuous*, with no strong pair-bonds. In others, mates form a relationship of some duration that is **monogamous** (one male mating with one female) or **polygamous** (an individual of one sex mating with several of the other). Polygamous relationships involve either *polygyny*, a single male and many females, or *polyandry*, a single female and multiple males. (Note that same-sex sexual behavior is also very common in many species.)

The extent to which males and females differ in appearance, a characteristic known as *sexual dimorphism*, typically varies with the type of mating system. Among monogamous species, males and females often look very similar. In contrast, among polygamous species, the sex that attracts multiple mating partners is typically showier and larger than the opposite sex **(Figure 51.14)**. We'll discuss the evolutionary basis of these differences shortly.

Figure 51.14

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    Relationship between mating system and male and female
forms.
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A comparison of relationship systems.

Parts are as follows.

Part a. Monogamy. In monogamous species, such as these western gulls (Larus occidentalis), males and females are often difficult to distinguish using external characteristics only.

Part b. Polygamy. Among polygynous species, such as elk (Cervus canadensis), the male (right) is often highly ornamented. Photo shows male with large antlers among 5 female elk.

Part c. Polyandry. In polyandrous species, such as these red-necked phalaropes (Phalaropus lobatus), females (right) are generally more ornamented than males. Photo shows female red-necked phalarope shown with stronger red marking on the neck and breast than her male mate.

Mating Systems and Parental Care

The needs of the young are an important factor constraining the evolution of mating systems. Most newly hatched birds, for instance, cannot care for themselves. Rather, they require a large, continuous food supply, a need that is difficult for a single parent to meet. In such cases, a male that stays with and helps a single mate may ultimately have more viable offspring than it would by going off to seek additional mates. This may explain why many bird species are monogamous. In contrast, for birds with young that can feed and care for themselves almost immediately after hatching, the males derive less benefit from staying with their partner. Males of these species, such as pheasants and quail, can maximize their reproductive success by seeking other mates, and polygyny is relatively common in such birds. In the case of mammals, the lactating female is often the only food source for the young, and males usually play no role in raising the young. In mammalian species where males protect the females and young, such as lions, a male or small group of males typically cares for a harem of many females.

Another factor influencing mating behavior and parental care is *certainty of paternity*. Young born to or eggs laid by a female definitely contain that female's genes. However, even if there is a strong pair-bond, a male other than the female's usual mate may on occasion produce offspring with that female. The certainty of paternity is relatively low in most species with internal fertilization because the acts of mating and birth (or mating and egg laying) are separated over time. This could be one reason that exclusively male parental care is rare in bird and mammal species. However, the males of many species with internal fertilization engage in behaviors that appear to increase their certainty of paternity. These behaviors include guarding females, removing any sperm from the female reproductive tract before copulation, and introducing large quantities of sperm that displace the sperm of other males.

Certainty of paternity is high when egg laying and mating occur together, as in external fertilization. This may explain why parental care in aquatic invertebrates, fishes, and amphibians, when it occurs at all, is at least as likely to be by males as by females (**Figure 51.15**; see also Figure 46.6). Among fishes and amphibians, parental care occurs in fewer than 10% of species with internal fertilization but in more than half of species with external fertilization.

Figure 51.15

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    Paternal care by a male yellowhead jawfish (*Opistognathus
aurifrons*).
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A male jawfish has a mouthful of eggs.

The male jawfish, which lives in tropical marine environments, holds the eggs it has fertilized in its mouth, keeping them aerated and protecting them from predators until the young fish hatch.

It is important to point out that certainty of paternity does not mean that animals are aware of those factors when they behave a certain way. Parental behavior correlated with certainty of paternity exists because it has been reinforced over generations by natural selection. The intriguing relationship between certainty of paternity and male parental care remains an area of active research.

Sexual Selection and Mate Choice

Sexual dimorphism results from sexual selection, a form of natural selection in which differences in reproductive success among individuals are a consequence of differences in mating success (see Concept 23.4). Sexual selection can take the form of *intersexual selection*, in which members of one sex choose mates on the basis of characteristics of the other sex, such as courtship songs, or *intrasexual selection*, which involves competition between members of one sex for mates.

Mate Choice by Females

Mate preferences of females may play a central role in the evolution of male behavior and anatomy through intersexual selection. Consider, for example, the courtship behavior of stalk-eyed flies. The eyes of these insects are at the tips of stalks, which are longer in males than in females. During courtship, a male approaches the female headfirst. Researchers have shown that females are more likely to mate with males that have relatively long eyestalks. Why would females favor this seemingly arbitrary trait? Ornaments such as long eyestalks in these flies and bright coloration in birds correlate in general with health and vitality. A female whose mate choice is a healthy male is likely to produce more offspring that survive to reproduce. As a result, males may compete with each other in ritualized contests to attract female attention **(Figure 51.16)**. In face-offs between male stalk-eyed flies, the male whose eyestalk length is smaller usually retreats peacefully.

Figure 51.16

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    A face-off between male stalk-eyed flies (*Cyrtodiopsis
whitei*) competing for female attention.
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Two male stalk-eyed flies face each other.

Mate choice can also be influenced by imprinting, as revealed by experiments carried out with zebra finches (*Taeniopygia guttata*). Both male and female zebra finches normally lack any feather crest on their head **(Figure 51.17)**. To explore whether parental appearance affects mate preference in offspring independent of any genetic influence, researchers provided zebra finches with artificial ornamentation. A 2.5-cm-long red feather was taped to the forehead feathers of either or both zebra finch parents when their chicks were 8 days old, approximately 2 days before they opened their eyes. A control group of zebra finches was raised by unadorned parents. When the chicks matured, they were presented with prospective mates that were either artificially ornamented with a red feather or non-ornamented **(Figure 51.18)**. Males showed no preference. Females raised by a male parent that was not ornamented also showed no preference. However, females raised by an ornamented male parent preferred ornamented males as their own mates. Thus, female finches apparently take cues from their male parents in choosing mates.

Figure 51.17

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    Appearance of zebra finches in nature.
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Two zebra finches stare at each other. They have different color patterns.

The male zebra finch (right) is more highly patterned and colorful than the female zebra finch (left).

Figure 51.18

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    Sexual selection influenced by imprinting.
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Sexual selection influenced by imprinting.

Experiments demonstrated that female zebra finch chicks that had imprinted on artificially ornamented fathers preferred ornamented males as adult mates. For all experimental groups, male offspring showed no preference for either ornamented or non-ornamented female mates.

Experimental groups of parental pairs of zebra finches include the following: both parents ornamented, males ornamented, and females ornamented. Control group: parents not ornamented. Male and female offspring are produced from each group. The mate preference for the female offspring for the experimental groups which include both parents ornamented and males ornamented was an ornamented male. For the experimental group where the females were ornamented and the control group, the mate preference of the female offspring was none.

Mate-choice copying, a behavior in which individuals in a population copy the mate choice of others, has been studied in the guppy, *Poecilia reticulata*. When a female guppy chooses between males with no other females present, the female almost always chooses the male with more orange coloration. To explore if the behavior of other females could influence this preference, an experiment was set up using both living females and artificial model females **(Figure 51.19)**. If a female guppy observed the model “courting” a male with less extensive orange markings, she often copied the preference of the model female. That is, the female chose the male that had been presented in association with a model female rather than a more orange alternative. The exceptions were also informative. Mate-choice copying typically did not change when the difference in coloration was particularly large; that is, the female guppy chose the male with the much stronger coloration even if a model female was associated with the less orange male. Mate-choice copying can thus mask genetically controlled female preference below a certain threshold of difference, in this case for male color.

Figure 51.19

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    Mate choice copying by female guppies (*Poecilia
reticulata*).
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Mate choice copying by female guppies.

In the absence of other females (control group), female guppies generally choose males with more orange coloration. However, when a female model is placed near one of the males (experimental group), female guppies often copy the apparent mate choice of the model, even if the male is less colorful than others. Guppy females ignored the mate choice of the model only if an alternative male had much more orange coloration.

Part a. Control sample: Female guppies prefer males with more orange coloration when presented with male guppies with varying degrees of coloration.

Part b. Experimental sample: Female guppies prefer males that are associated with another female when a female model in introduced in mock courtship with the less orange male guppy.

Mate-choice copying, a form of social learning, has also been observed in several other fish and bird species. What is the selective pressure for such a mechanism? One possibility is that a female that mates with males that are attractive to other females increases the probability that her male offspring will also be attractive and have high reproductive success.

Male Competition for Mates

The previous examples show how female choice can select for one best type of male in a given situation, resulting in low variation among males. Similarly, male competition for mates can reduce variation among males. Such competition may involve *agonistic behavior*, a contest, typically ritualized, that determines which competitor gains access to a resource, such as food or a mate **(**Figure 51.20**; see also Figure 51.16).

Figure 51.20

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Agonistic interaction.
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Male eastern grey kangaroos boxing.

Male eastern grey kangaroos (Macropus giganteus) often "box" in contests that determine which male is most likely to mate with an available female. Typically, one male snorts loudly and strikes the other with his forelimbs. If the male under attack does not retreat, the fight may escalate into grappling or the two males balancing on their tails while attempting to kick each other with the sharp toenails of their hind feet.

Video

Agonistic Behavior in Wolves

image

Play

image

Play

Video

Snake Ritual Wrestling

image

Play

image

Play

Despite the potential for male competition to select for reduced variation, behavioral and morphological variation in males is extremely high in some vertebrate species, including species of fish and deer, as well as in a wide variety of invertebrates. In some species, sexual selection has led to the evolution of alternative male mating behavior and morphology. How do scientists analyze situations where more than one mating behavior can result in successful reproduction? One approach relies on the rules that govern games.

Applying Game Theory

Often, the fitness of a particular behavioral phenotype is influenced by other behavioral phenotypes in the population. In studying such situations, behavioral ecologists use a range of tools, including game theory. Developed by American mathematician John Nash and others to model human economic behavior, **game theory** evaluates alternative strategies in situations where the outcome depends on the strategies of all the individuals involved.

As an example of applying game theory to mating behavior, let's consider the common side-blotched lizard (*Uta stansburiana*) of California. Genetic variations give rise to males with orange, blue, or yellow throats **(Figure 51.21)**. One would expect that natural selection would favor one of the three color types, yet all three persist. Why? The answer appears to lie in the fact that each throat color is associated with a different pattern of behavior: Orange-throat males are the most aggressive and defend large territories that contain many females. Blue-throat males are also territorial but defend smaller territories and fewer females. Yellow-throats are nonterritorial males that mimic females and use "sneaky" tactics to gain the chance to mate.

Figure 51.21

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    Male polymorphism in the common side-blotched lizard (*Uta
stansburiana*).
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A person holds 3 lizards with different colors.

An orange-throat male, left; a blue-throat male, center; a yellow-throat male, right.

Evidence indicates that the mating success of each male lizard type is influenced by the relative abundance of the other types, an example of frequency-dependent selection. In one study population, the most frequent throat coloration changed over a period of several years from blue to orange to yellow and back to blue.

By comparing the competition between common side-blotched lizard males to the children's game of rock-paper-scissors, scientists devised an explanation for the cycles of variation in the lizard population. In the game, paper defeats rock, rock defeats scissors, and scissors defeats paper. Each hand symbol thus wins one matchup but loses the other. Similarly, each type of male lizard has an advantage over one of the other two types:

- When blue-throats are abundant, they can defend the few females in their territories from the advances of the sneaky yellow-throat males. However, blue-throats cannot defend their territories against the hyperaggressive orange-throats.
- Once the orange-throats become the most abundant, the larger number of females in each territory provides the opportunity for the yellow-throats to have greater mating success.
- The yellow-throats become more frequent, but then give way to the blue-throats, whose tactic of guarding small territories once again allows them the most success.

Thus, following the population over time, one sees a persistence of all three color types and a periodic shift in which type is most prevalent.

Game theory provides a way to think about complex evolutionary problems in which relative performance (reproductive success relative to other phenotypes), not absolute performance, is the key to understanding the evolution of behavior. This makes game theory an important tool because the relative performance of one phenotype compared with others is a measure of Darwinian fitness.